

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed **AI-assisted documentation services** using **Python** and **LLM APIs** to automate content classification, tagging, and summarization across enterprise technical documentation, reducing manual documentation review time by **40%** and enabling writers to focus on high-value content creation.
- Designed and implemented **DITA XML transformation pipelines** and **metadata enrichment models** using **Python** and **SQL** to structure enterprise content for AI consumption, enabling reliable chunking, semantic indexing, and RAG pipeline integration across **50,000+** documentation artifacts.
- Built **AI-powered content quality APIs** and integrated them into **Git-based CI/CD workflows (Azure DevOps)** to validate documentation structure, enforce metadata standards, and flag consistency issues pre-publication, improving content quality gates by **35%** without disrupting author workflows.
- Architected **LLM-ready content models** and **data pipeline transformations** to prepare **DITA-structured documentation** for AI agents, defining schema-driven enrichment strategies (semantic tags, context links, source attribution) that powered downstream search and Q&A assistant experiences.
- Integrated **LLM APIs** and **RAG pipelines** with Bentley's content authoring stack (**Oxygen/DITA, Git**) to deliver intelligent documentation assistance—link validation, content summarization, and automated structure checks—achieving **15+ hours/week** of saved editorial labor across **8 documentation teams**.
- Collaborated with documentation leaders, platform engineers, and product stakeholders to identify high-impact AI automation use cases, translating business requirements into technical architecture for AI-assisted workflows and contributing to Documentation Community of Practice standards for AI-ready content preparation.

SKILLS

AI & LLM Engineering – LLM API integration, RAG pipelines, prompt engineering, content enrichment for AI systems, AI-assisted workflow automation.

Structured Content & Documentation Platforms – DITA XML, schema-driven data models, content metadata, Oxygen Editor, OpenDITA, technical documentation standards.

Backend & Data Pipeline Development – Python, TypeScript/JavaScript, REST APIs, Git-based workflows, Azure DevOps, CI/CD pipeline integration, microservices.

Data Engineering & Transformations – SQL (PostgreSQL, MySQL, Oracle), ETL pipelines, data modelling, content chunking and indexing strategies, semantic tagging.

Cloud & Infrastructure – AWS (EC2, S3, Lambda), Docker, Linux, Jenkins, Git, environment automation for content systems.

Collaboration & Requirements – Cross-functional partnership with technical writers, platform engineers, product teams, stakeholder analysis, documentation standards definition.

Analytical & Visualization – Power BI, Tableau, data-driven decision-making, content quality metrics, performance measurement.

EDUCATION AND TRAINING

PROJECTS

AI-Ready Documentation Transformation & LLM Integration Platform (HCL Technologies)

- Led end-to-end design and delivery of a **DITA XML-to-AI transformation pipeline** using **Python**, **LLM APIs** (GPT-4 and Claude), and **RAG frameworks** to restructure **100,000+ pages** of enterprise technical documentation into AI-consumable formats with semantic metadata, context enrichment, and source attribution.
- Built **intelligent content quality services** using **Python FastAPI** and **LLM-powered validation** that automatically detect structural inconsistencies, missing cross-references, metadata gaps, and tone/style violations in **DITA XML** documents, integrated into **Azure DevOps CI/CD pipelines** to enforce quality gates pre-publication with **zero friction** for writers.
- Designed a **metadata and context enrichment model** using **SQL**, **graph-based semantic linking**, and **LLM-generated summaries** that enabled downstream AI agents (search assistants, Q&A systems, code-example finders) to retrieve and reason over documentation with **92% relevance accuracy**, reducing user query resolution time by **55%**.
- Implemented **AI-assisted authoring tools** (content classification, intelligent tagging, link suggestions, automated summarization) via **REST APIs** and **Oxygen Editor plug-ins**, reducing average documentation creation time by **30 hours/month** across **12 writer teams** while maintaining editorial control and voice.
- Established technical standards and best practices for **AI-ready documentation** through collaboration with documentation leadership, architects, and the Documentation Community of Practice, defining structure, semantics, metadata schemas, and context models that became the enterprise standard for AI consumption across all product lines.

SWIFT Payments Operations Intelligence & Regulatory Compliance Dashboard (HCL Technologies)

- Architected and deployed a real-time **payments operations monitoring platform** using **Python ETL pipelines**, **PostgreSQL**, and **Power BI dashboards** to ingest **SWIFT MT/MX message flows** and provide transaction-flow visibility, SLA tracking, and operational alerting for a Tier-1 banking client processing **50,000+ daily payment messages**.
- Designed **ISO 20022 message transformation and enrichment logic** using **SQL** and **Python** to normalize SEPA, cross-border, and instant payment data for BI consumption, enabling regulatory reporting (**GDPR** compliance, audit trails) and reconciliation workflows with **99.8% data accuracy**.
- Reduced P1/P2 payment incident mean-time-to-resolution (MTTR) by **65%** through intelligent **Dynatrace** and **CloudWatch** alerting, correlation rules, and **Power BI operational dashboards** that surfaced transaction anomalies, bottlenecks, and regulatory edge cases in real time across interconnected ERP and payments systems.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)